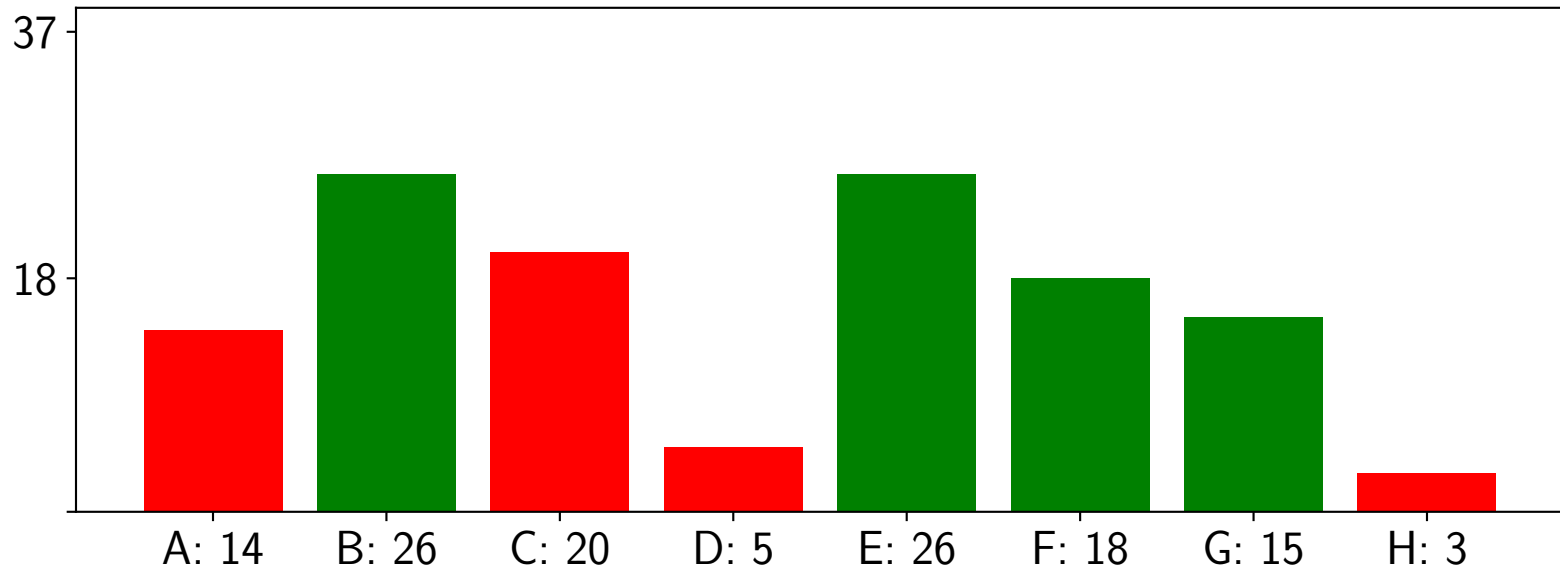
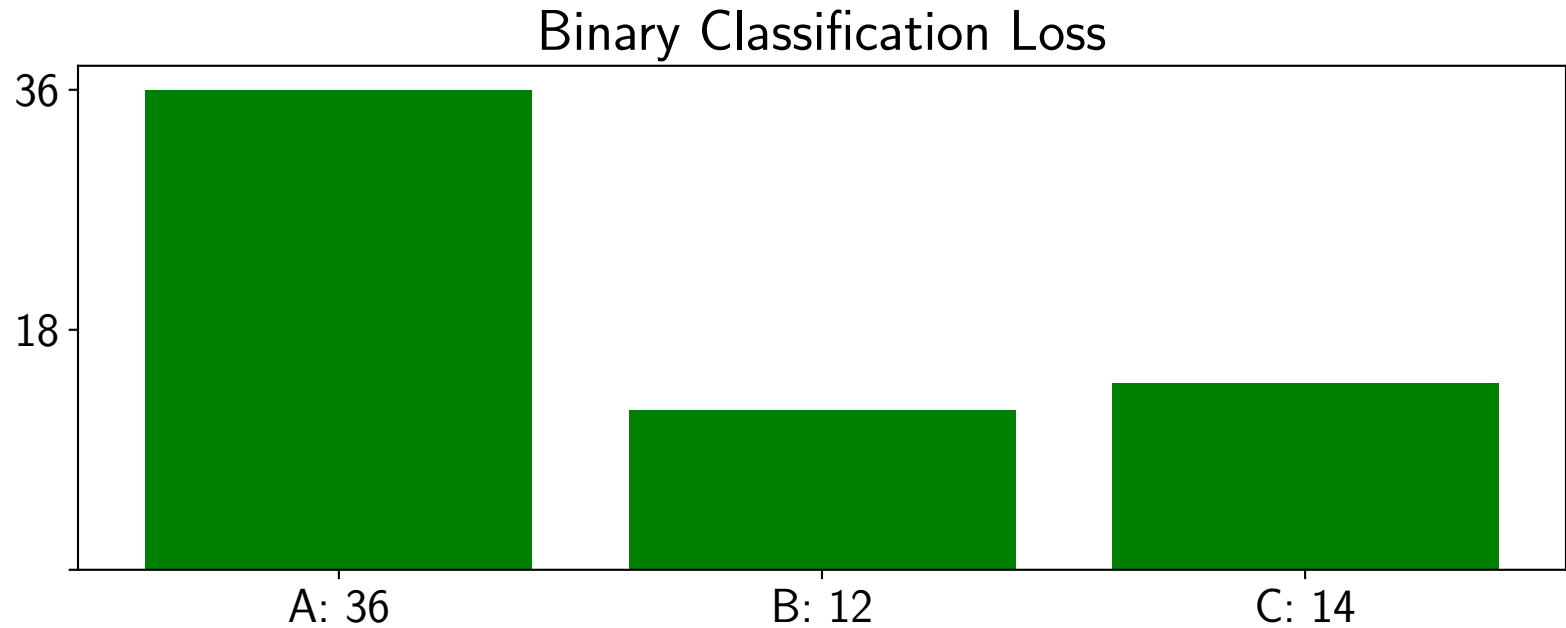


Classification



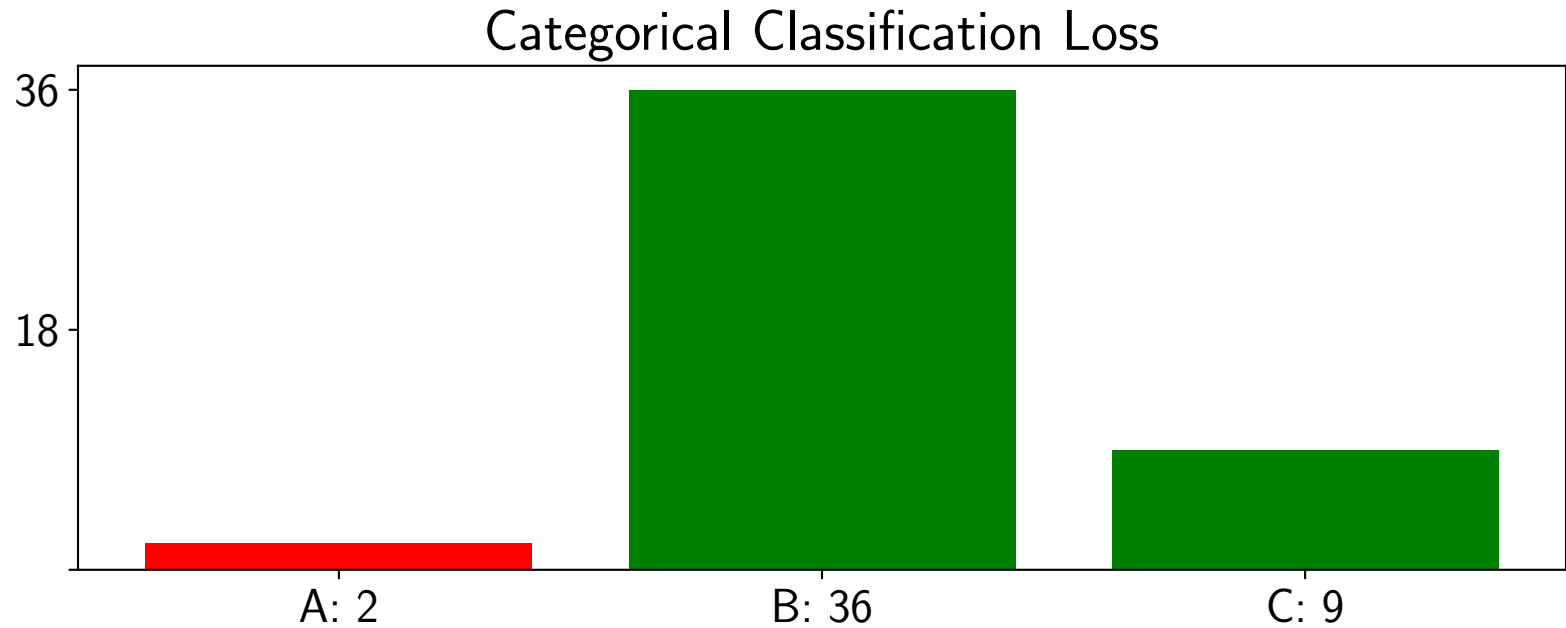
Which of these statements are correct?

- A Binary Cross-Entropy Loss pairs well with $\tanh$ activation function.
- B Categorical Cross-Entropy Loss pairs well with SoftMax activation function.
- C The higher the logit values, the higher the SoftMax probability.
- D For a two-class classification, a network with two outputs is required.
- E For a three-class classification, a network with three outputs is required.
- F The derivatives of Binary Cross-Entropy loss on logistic output activation and Euclidean loss on linear output activation are identical.
- G The derivatives of Categorical Cross-Entropy loss on SoftMax output activation and Euclidean loss on linear output activation are identical.
- H The model with the best training set accuracy should be preferred.



Which loss function can be used for binary classification in deep networks?

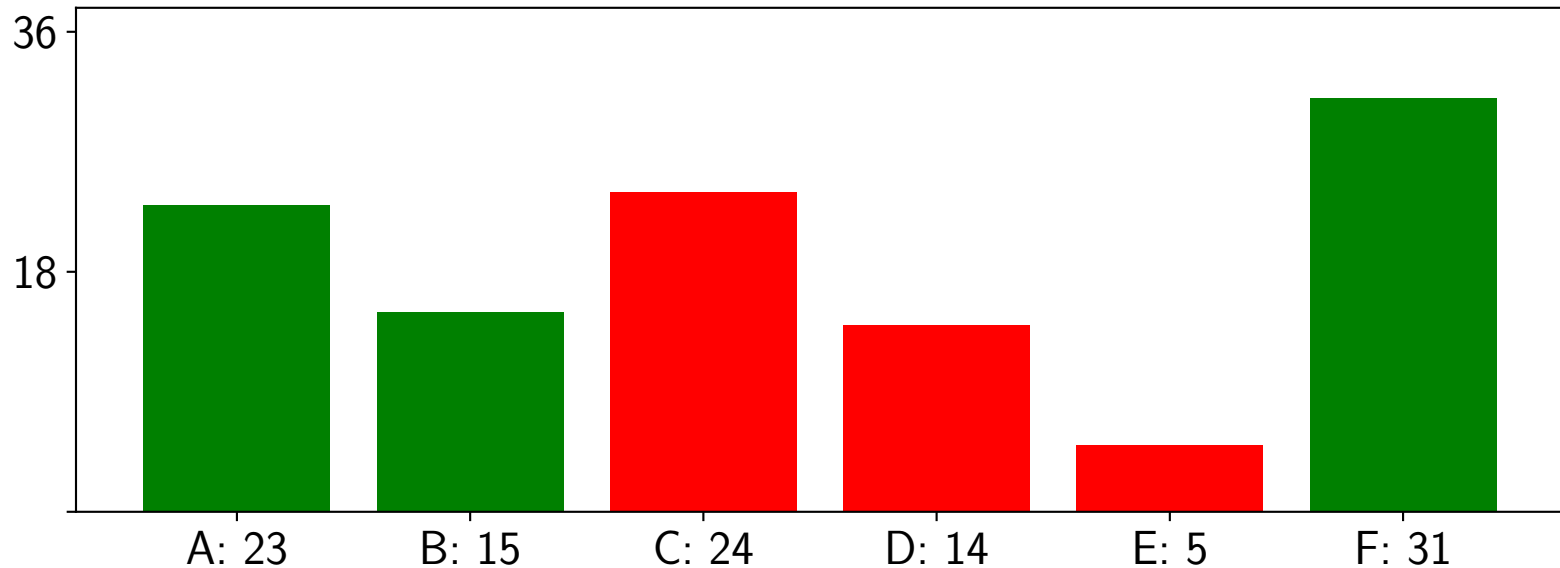
- A Binary Cross-Entropy Loss
- B Categorical Cross-Entropy Loss
- C Euclidean Loss



Which loss function can be used for categorical classification in deep networks?

- A Binary Cross-Entropy Loss
- B Categorical Cross-Entropy Loss
- C Euclidean Loss

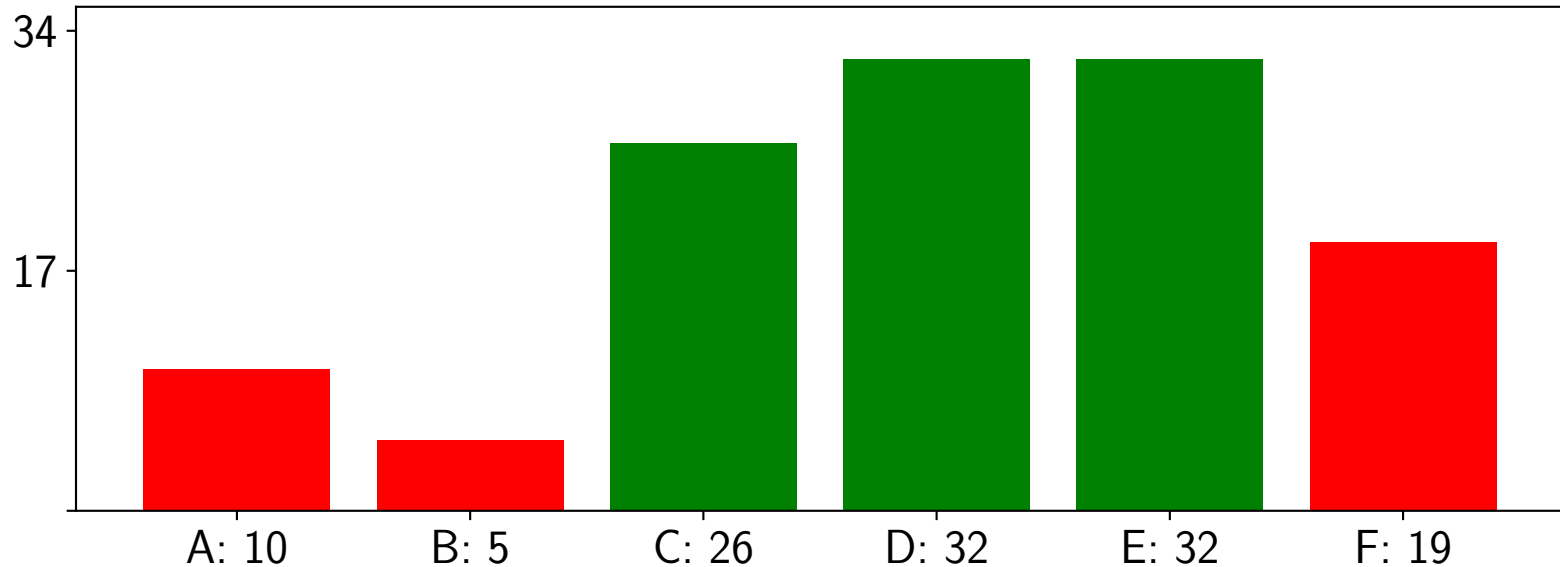
Automatic Differentiation



Mark all correct statements.

- A An execution graph is used to store operations on data.
- B Each operation remembers its input variables.
- C For any function, partial derivatives w.r.t. all its inputs are computed.
- D In PyTorch, the backward function can be applied to any variable of type `torch.tensor`.
- E For vectorized automatic differentiation, Hessian matrices are computed.
- F Automatic differentiation can be applied to any differentiable function with finite numbers of inputs.

Convolutional Kernels



What do you know about convolutional kernels?

- A The number of outputs of a convolutional operation can be selected freely.
- B A 3×3 convolutional kernel has 9 learnable parameters.
- C Zero padding can be used to keep input and output size identical.
- D Striding and Pooling are two methods to reduce the size of the convolution output.
- E Commonly applied pooling operations are maximum pooling and average pooling.
- F The weights of a channeled 2D convolution build a 3D tensor of learnable parameters.